

Assignment :

JavaScript Basics Practice – Operators, Conditionals, Loops, and Functions

Assignment Instructions:

1. This assignment is designed to help you practice **basic JavaScript concepts**, including:
 - Operators
 - Conditional statements (if, else, else if)
 - Loops (for, while)
 - Functions
2. Read each exercise carefully before writing your code.
3. Open new folder called “**Javascript_Ass1**” and open it in Vs_code then create file call “**JavaScriptBasics.js**” then write you code answer inside.

N:B Include the comment of show question number like *// Exercise 1: Addition Function*

◆ Exercise 1: Addition Function

Write a JavaScript function called **add** that takes **two numbers** as input.

Your task:

1. Create a function named add
2. The function should accept two parameters (for example: a and b)
3. Inside the function, calculate the **sum** of the two numbers
4. Return the result
5. Call the function and print the result using console.log()

Example Input: a = 5, b = 3

Expected Output: The sum is: 8

◆ Exercise 2: Check Even or Odd

Write a program that checks whether a number is **even or odd**.

Your task:

1. Create a variable and assign it a number
2. Use the modulus operator % to check:
 - If the number is divisible by 2 → Even
 - Otherwise → Odd
3. Print the result

Example Input: number = 7

Expected Output: 7 is an odd number

◆ Exercise 3: Find the Largest Number

Write a program to find the **largest of two numbers**.

Your task:

1. Create two variables
2. Compare them using if...else
3. Print which number is larger, or if they are equal

Example Input: a = 10 , b = 20

Expected Output: 20 is the largest number

◆ Exercise 4: Student Grade Checker

Write a program that assigns a grade based on marks.

Your task:

1. Create a variable marks
2. Use if...else if...else to assign grade:
 - 80–100 → A
 - 70–79 → B
 - 60–69 → C
 - 50–59 → D
 - Below 50 → F
3. Print the grade

Example Input: marks = 75

Expected Output: Grade: B

◆ Exercise 5: Print Numbers from 1 to 10

Write a program using a loop.

Your task:

1. Use a for loop
2. Print numbers from 1 up to 10

Expected Output:

1
....
10

◆ Exercise 6: Sum of Numbers from 1 to 10

Write a program to calculate the total sum.

Your task:

1. Use a loop
2. Add numbers from 1 to 10
3. Store the result in a variable
4. Print the total sum

Expected Output: The sum is: 5050

◆ **Exercise 7: Multiplication Table**

Write a program to display a multiplication table.

Your task:

1. Create a variable (example: num = 5)
2. Use a loop from 1 to 10
3. Print multiplication results

Example Input: num = 5

Expected Output:

5 x 1 = 5

5 x 10 = 50

◆ **Exercise 8: Simple Login System**

Write a program to simulate login.

Your task:

1. Create variables:
 - username
 - password
2. Check:
 - If username = "admin" AND password = "1234"
3. Print:
 - "Login successful" OR "Invalid credentials"

◆ Exercise 10: Function to Find Maximum of 3 Numbers

Write a function that finds the largest number.

Your task:

1. Create a function named findMax
2. It should accept 3 numbers
3. Return the largest number
4. Call the function and print the result

Example Input:10, 25, 15

Expected Output: The largest number is: 25

◆ Exercise 11: Factorial Function

Write a function to calculate factorial.

Your task:

1. Create a function named factorial
2. Use a loop to calculate factorial
3. Return the result

Example Input: 5

Expected Output: 120

Exercise 12: Simple Discount System

Write a JavaScript program that calculates the **final price after applying a discount**.

Your task:

1. Create a variable called price and assign it a value
2. Use an if...else statement to check:
 - If the price is **greater than 100**
 - Apply a **10% discount**
 - Otherwise
 - No discount is applied
3. Calculate the final price after discount (if any)
4. Print the result using **console.log()**

Example Input: let price = 150;

Expected Output: Discount applied: 10%

Final price: 135